

Zhexiang (Bruce) Li

M.S. Student in Computer Science, UCLA

bruceli@ucla.edu +1 831-239-2728 Los Angeles, CA [Website](#) [GitHub](#) [LinkedIn](#)

EDUCATION

University of California, Los Angeles Sep 2026 - Jun 2028 (expected)

M.S. in Computer Science

University of Southern California May 2026

B.S. in Computer Science | GPA 3.76 / 4.00 | Viterbi Dean's List, Fall 2024-Spring 2026

TECHNICAL SKILLS

Languages Python, C/C++, JavaScript/TypeScript, SQL

ML and scientific computing PyTorch, TensorFlow, Triton, CUDA, RDKit, Chemprop, recommender systems, multimodal inference

Software and data FastAPI, React, Next.js, PostgreSQL, Supabase, Docker, Linux

RESEARCH EXPERIENCE

Health Intelligence Lab (HAIL) | UCLA Current

Research with Prof. Yuzhe Yang

Working on foundation models for type 1 diabetes (T1D).

Machine Learning for Antiviral Drug Discovery | USC Spring 2026

Research with Prof. Yan Liu's group

- Benchmarked RDKit, Chemprop, and hybrid models on seven TDC datasets with scaffold splits; ran feature-importance and SHAP analyses.

Efficient Attention via Pre-Scoring | First author 2024 - 2025

Advised by Prof. David P. Woodruff, Carnegie Mellon University | [arXiv:2505.11040](https://arxiv.org/abs/2505.11040)

- Proposed pre-scoring to select informative key intervals before attention; implemented Triton kernels and benchmarked against HyperAttention and LevAttention.

INDUSTRY EXPERIENCE

AI Engineer Intern | PMAOS / Next Big Seek Jul - Aug 2026

- Designed a short-video recommender with bidirectional user-video interest propagation and separate session, bridge, and long-term exposure budgets. Added fatigue controls that cap repeated topics while reserving traffic for new interests.
- Built 11-language audio-visual affect inference with synchronization checks, presentation-attack detection, evidence fusion, and a fixed-point Q15 decision layer that can abstain or flag modality conflicts.
- Validated the recommendation logic with 629 tests and the multimodal pipeline with 139 Python tests, type and contract checks, and cross-compilation for Cortex-M3 and ARM926EJ-S.
- Evaluated a 9 mm piezoelectric millirobot concept and designed a PZT-steel cantilever experiment with acceptance criteria; work remained at the experimental planning stage.

Full-Stack Engineer | Terlingua Threads Jan - May 2026

- Built Shopify-integrated pages and Google Maps features; improved responsive layouts, accessibility, and SEO.

Co-Founder and CTO | CrAlg Capital 2025 - Mar 2026

- Built the data and workflow architecture for an AI-assisted housing debt-collection platform; set technical priorities and delivery standards.

AI Engineer Intern | Piezolab Medical

Jun - Aug 2025

- Built an LLM-assisted pipeline to ingest, clean, and cluster 5,000+ medical-malpractice documents for quality-of-care review.
- Wrote experiment checklists and acceptance tests for vascular-care laboratory workflows.

Application Engineer Intern | Synaptics

Jun - Aug 2023

- Validated product firmware and built an ESP32-to-Alibaba Cloud pipeline that published I2C sensor data as MQTT topics for internal dashboards.

PUBLICATIONS AND MANUSCRIPTS

MedTokenBudget: Lesion-Preserving Token Routing for Dermoscopic Image Classification

Summer 2026

Lesion-aware token routing under fixed budgets, trained with mask supervision and evaluated for lesion retention on ISIC 2019. Preprint; [arXiv:2609.30613](https://arxiv.org/abs/2609.30613).

Mass-Conservation Token Transformers for Auditable MS/MS Spectrum Retrieval

Summer 2026

Encoded mass-conserving fragment pairs as sparse tokens for spectrum reranking; tested ranking dependence with hard negatives and token-removal experiments. Submitted to AAAI 2027; preprint forthcoming.

Efficient Attention via Pre-Scoring

2025

First author; preprint. [arXiv:2505.11040](https://arxiv.org/abs/2505.11040)

Game Theory in Companies

2023

Modeled two-player release decisions and built a Python + Gambit pipeline to enumerate Nash equilibria. [DOI](#)

SELECTED PROJECTS

Fridge Manager

2026

Built a Next.js and Supabase meal planner that uses inventory, leftovers, recipes, dietary preferences, and available time to produce meal and grocery plans.

AI Business Card Hub

2026 - Present

Built an iOS organizer with vision parsing, OCR fallback, multi-photo review, structured people and company records, search, and bilingual display. [Code](#)

ADDITIONAL EXPERIENCE

Investment Analyst Intern | GUDU Medical

Jan 2021 - Present

- Screen MedTech opportunities and support market research, due diligence, financial modeling, and valuation.